La Scuola d'Italia / Dr. Huson / IB Math: Quiz
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4.19 Quiz: Probability (Algebra)

1. Events A and B are independent. It is given that $\mathrm{P}(A) = 2\,\mathrm{P}(B)$ and $\mathrm{P}(A \cup B) = 0.72$.

   Let $x = \mathrm{P}(B)$.

   (a) Show that $2x^2 - 3x + 0.72 = 0$. [3]

   (b) Hence find $\mathrm{P}(B)$. [2]

   (c) Find $\mathrm{P}(A \cap B)$. [1]

a) $P(A) = 2x$

INDEPENDENT $\Rightarrow$ $P(A)P(B) = P(A \cap B)$

$P(A \cap B) = 2x \cdot x = 2x^2$

$P(A \cup B) = 2x + x - 2x^2 = 0.72$

$2x^2 - 3x + 0.72 = 0$

b) $x = \dfrac{-(-3) \pm \sqrt{(-3)^2 - 4(2)(0.72)}}{2(2)}$

$= 0.3$ disregard $x > 1$

c) $P(A) = 2 \cdot 0.3 = 0.6$

$P(A \cap B) = 0.6 \cdot 0.3$

$= 0.18$

2. In a school, 80 students were surveyed about activities A and B.

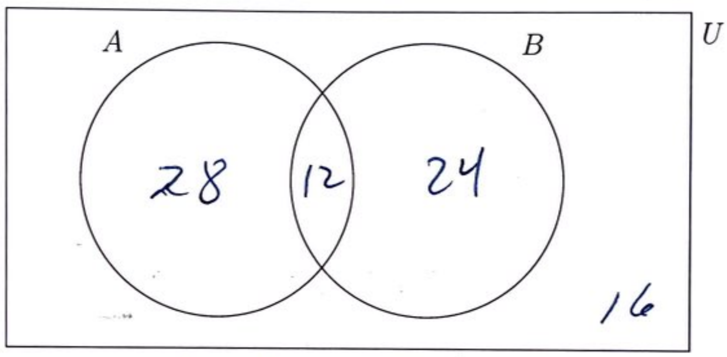
- $P(A) = 0.5$
- $P(B \mid A) = 0.3$
- $P(A' \cap B') = 0.2$

(a) Find $P(A \cap B)$. [2]

(b) Find $P(B)$. [2]

(c) Complete the Venn diagram below, showing the number of students in each region. [2]

(d) Determine whether events A and B are independent. Justify your answer. [2]



a) $P(B|A) = \frac{P(A\cap B)}{P(A)} = \frac{P(A\cap B)}{0.5} = 0.3$

$P(A\cap B) = 0.15$

b) $P(A\cup B) = P(A) + P(B) - P(A\cap B) = 1 - 0.2$

$0.5 + P(B) - 0.15 = 0.8$

$P(B) = 0.45$

c) $0.15 \cdot 80 = 12$

$0.5 \cdot 80 - 12 = 28$

$0.45 \cdot 80 - 12 = \cancel{36}\ 24$

d) $P(A)P(B) = 0.5 \cdot 0.45 = 0.225$

$P(A\cap B) = 0.15$

$0.225 \neq 0.15$ NOT Independent